

Effect of Mass Distribution of Handheld Proxy on Redirection Detection Thresholds, Sense of Agency and Workload in Virtual Reality

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ABSTRACT

Handheld physical proxies when appropriately designed and remapped to virtual objects in virtual reality (VR) can support various purposeful motion-based applications such as virtual rehabilitation and exergames. Mass distribution, which is a major ergonomic factor in handheld proxies, may vary and influence user experiences. This study investigates how mass distribution of a handheld stick affects users' perceptual abilities when performing redirected pointing task in VR. Results showed that mass distribution significantly influences redirection detection threshold for slow gain. When internal weights were designed close to hand, users experienced the highest sense of agency, the lowest physical and mental workload.

Index Terms: Redirection detection thresholds, handheld physical proxies, sense of agency.

1 INTRODUCTION

Handheld proxies, when appropriately designed and remapped to virtual objects in virtual environments, can support purposeful motion-based virtual reality (VR) applications such as virtual rehabilitation, exergaming, and training [6]. As virtual rehabilitation increasingly incorporates physical rehabilitation tools, designing a tool's passive haptic properties (e.g., mass, mass distribution, and shape) becomes critical for shaping users' perceptual experience. Mass distribution is a key passive haptic property that changes a handheld tool's rotational inertia and the effort required for control. Variations in center of mass of proxy may affect perception of visuomotor mismatch and influence a series of experiences relevant to motor control, such sense of agency (SoA), and the mental and physical resources required [5].

Motion redirection, which refers to a technique by introducing a spatial displacement between users' virtual movements and physical movements, has been widely used in motion-based VR interactions to enhance user performance and haptic experience [2, 4, 5, 7]. Maintaining a good level of SoA, mental workload and perceived exertion is highly essential for VR programs based on motion redirection. Exploring the detectable ranges of thresholds and controlling experiences influenced mass distribution of handheld tools will help to form design guidelines of VR motion-based applications that involve physical proxies.

In the literature reviewed, the weight design of VR proxies was found to have substantial influence on user's perceptions of virtual objects [6]. Previous studies have explored the influence of task complexity and types of redirection in both virtual hand reaching [1, 2, 7] and walking redirection [4]. However, the effects of mass distribution on user experience, particular for redirection detection thresholds of proxies has not been fully explored.

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This study aims to estimate detection threshold and explore user experience influenced by redirected handheld stick with different mass distribution. The study results contribute to the design of physical tool for health-related VR application, such as virtual rehabilitation and exergames.

2 METHOD

An experiment was designed to investigate the effect of mass distribution of a redirected handheld stick on users' redirection detection threshold, SoA, and perceived exertion and cognitive load in VR.

2.1 Handheld Tool with Different Mass Distributions

This study uses a stick-shaped physical proxy with a length of 50 cm and a total weight of 400 g, which can be applied to motion-based interaction scenarios in VR, such as virtual exergames, rehabilitation, and industrial training [6].

Three mass distribution conditions were created by embedding a 230 g weight at different positions inside the stick, as shown in Figure 1(a), corresponding to **proximal**, **middle**, and **distal** mass distribution conditions. An HTC Vive Tracker 3.0 was fixed to the grasping end of the stick to track motion data.

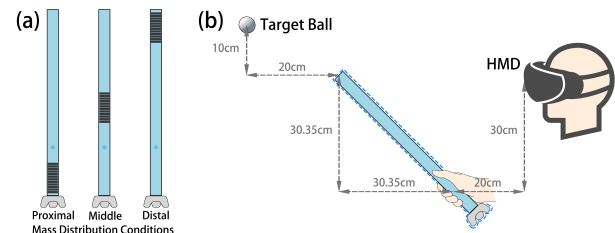


Figure 1: Handheld tool mass distribution (a) and the redirected pointing task (b).

2.2 Experiment Design

2.2.1 Participants

A total of 30 participants (17 males, 13 females; aged 18-25, $M = 22.10$, $SD = 1.86$) took part in the study. All participants provided written informed consent and reported being right-handed with limited prior VR experience.

2.2.2 Task Design and Procedure

As shown in Figure 1(b), participants performed a point-and-return task commonly used in gaming interfaces, in which they controlled a virtual stick using its physical counterpart to collide with a target sphere, positioned at a fixed distance from the calibration position (dashed lines, as shown in Figure 1(b)). The virtual stick was applied with a horizontal warping technique for redirection [7], where its virtual position was offset from its tracked position in the physical world. When participants reached the calibration position, a beep sound was played to indicate the onset of redirection and a trial begins. Once they reached the target sphere and returned to the

calibration position, a trial finished. The virtual environment was developed by Unity and was streamed via an HTC Vive pro headset.

Preparation session. Participants are required to practice the redirected point-and-return task and gain an understanding with *normal* and *nor normal* virtual movements. This session involves 21 trials (informed motion gain gradually applied from 0.5 to 2.0.) to control the proxy under proximal mass distribution condition.

Experimental session. This session follows a within-subject design and each participants completed all conditions and trials. The experiment includes three mass distribution conditions and 21 motion gain: 10 *slow* gain ($G = 0.50, 0.55, 0.60, \dots, 0.90, 0.95$), 10 *fast* gain ($G = 1.1, 1.2, 1.3, \dots, 1.9, 2.0$), and one *normal* value ($G = 1.0$). Each combination of condition and motion gain was repeated twice, with a total of 126 trials ($21 \times 2 \times 3$) included. After performing all the trials in a condition, participants are required to complete a questionnaire. The order of the three conditions (using Latin square order design) and the 42 trials (using randperm in Matlab) were all randomized for each participant. This process continued until all three conditions were completed.

Measures. This study employed a psychophysical forced-choice design to estimate the absolute detection threshold of redirection [2, 3]. After completing each trial, an interface appears for participants to judge whether they felt the movements of the virtual stick were *normal* (defined in the instructions as exactly matching their own movements) once they finished a trial [3]. Participants have to answer *yes (normal)* or *no (not normal)* according to their perceptions of the movements by clicking the button. Detection thresholds were derived by fitting psychometric functions to the proportion of *yes (normal)* responses across motion gains, with the threshold defined at 50% response probability. In psychophysical research, a 50% probability level is often employed to find the absolute threshold, i.e., the lowest intensity of a stimulus that can be detected [1, 2]. To measure of user experience influenced by mass distribution, participants are required to self-report their SoA, perceived exertion (RPE), and cognitive workload (NASA-TLX) on a 7-point Likert scale (-3 to +3) once they finished a condition. In this study, higher ratings of average SoA and RPE refers to stronger feeling of SoA and perceived exertion, while the higher ratings of NASA-TLX results indicates lower mental workload.

3 RESULTS

Figure 2 shows the results of threshold fitting for the probability of a *yes(normal)* response against the fast and slow gains in the experiment, estimated using the *quickpsy* package in R.

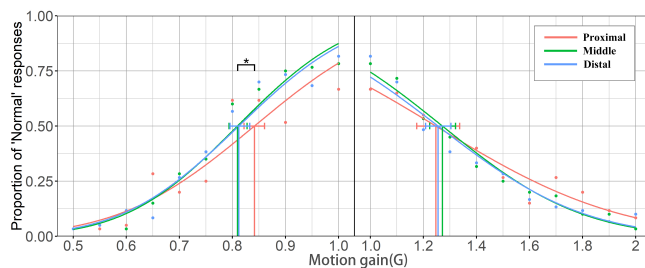


Figure 2: Fitted psychometric functions for motion gains $G < 1$ (left) and $G > 1$ (right). Dashed lines indicate the 50% detection thresholds. Error bars denote 95% confidence intervals.

For *slow* gains ($G < 1$), **proximal** condition exhibited the highest 50% detection threshold ($thre = 0.841$), followed by **distal** condition ($thre = 0.812$) and **middle** condition ($thre = 0.810$). Pairwise comparisons (using *thresholdcomparisons* function in *quickpsy*) showed that the estimated redirection detection threshold of proximal condition was significantly higher than that of middle condition

($p < .05$) in *slow* gains. For *fast* gains ($G > 1$), significant effect of mass distribution on detection threshold was not found (proximal: $thre = 1.247$; middle: $thre = 1.272$; distal: $thre = 1.255$, $p > .05$). Therefore, the **proximal** condition generated the smallest estimated range of detection threshold (from 0.841 to 1.255).

Table 1 shows the self-reported results of SoA, RPE, and NASA-TLX. Repeated-measures ANOVAs revealed significant effects of mass distribution on all three self-ratings. The proximal mass distribution condition showed the highest ratings of SoA and the lowest perceived physical and mental workload, as measured by RPE and NASA-TLX.

Table 1: Results of Self-ratings of SoA, RPE and NASA-TLX

Measure	Statistics	Mass distribution			p
		Proximal	Middle	Distal	
SoA	M	2.48	2.29	2.00	0.026
	SD	1.04	1.13	1.16	
RPE	M	1.10	1.42	2.47	<0.001
	SD	0.86	1.02	1.51	
NASA-TLX	M	1.57	1.33	0.72	<0.001
	SD	0.96	0.83	1.29	

4 CONCLUSION

This study examined the impact of a handheld stick's mass distribution on user perception during a redirected pointing task in VR. The findings indicate that mass distribution plays a significant role in influencing users' detection threshold for redirection, particularly at slow motion gains. The strongest SoA and the lowest physical and mental workload were reported when the internal weights were positioned closer to the hand. The findings in this study contributes to the ergonomic guidance of handheld tool for VR applications to support natural and controllable user experience.

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